31 · Making sperm and eggs

The words of this track, as it says them. Bold brown is the examiner.

Making sperm and eggs. Four marks.

Describe the gamete production process in males and in females. What hormone from the brain drives this?

The hook. Sperm start at puberty, never stop, and one cell makes four. Eggs start before birth, pause, go one a month, finish only if a sperm gets in, and one cell makes one.

Males first: spermatogenesis. It happens in the seminiferous tubules of the testes, continuously, from puberty on. Spermatogonia divide by mitosis into primary spermatocytes. Meiosis one gives secondary spermatocytes. Meiosis two gives spermatids. Then comes spermiogenesis: they lose most of their cytoplasm and grow a tail. One primary spermatocyte, four sperm.

Females: oogenesis. It starts in fetal life. Oogonia divide by mitosis into primary oocytes, which begin meiosis one, and then stop, arrested in prophase one, until puberty. There are about two million of them at birth. From puberty, each month one primary oocyte finishes meiosis one, giving a secondary oocyte and the first polar body. The secondary oocyte arrests again, in metaphase two, and that's what is ovulated. Meiosis two is only completed if a sperm enters it, giving one ovum and a second polar body. And egg production ends at menopause.

The hormone: FSH, follicle-stimulating hormone, from the anterior pituitary, working together with LH.

The traps. One primary spermatocyte makes four sperm. One primary oocyte makes only one egg; the rest are polar bodies. Second: eggs start before birth, and sperm start at puberty. Third: the egg only finishes meiosis two after a sperm gets in.

Once more. Making sperm and eggs.

Spermatogenesis happens in the seminiferous tubules, continuously from puberty. Spermatogonia divide by mitosis into primary spermatocytes, then meiosis one and two give spermatids, which become four sperm. Oogenesis starts in fetal life: primary oocytes arrest in prophase one until puberty. From puberty, one a month completes meiosis one and is ovulated, and meiosis two finishes only if a sperm enters, giving one ovum. Both are driven by FSH from the anterior pituitary, with LH.